

SOPHIE YU

2400 Sheridan Road, Evanston, IL, 60201 | (773) 290-4492 | sophieyu1028@gmail.com

<https://www.linkedin.com/in/sophie-y-61275829a/>

Summary

Electrical Engineering student at Northwestern University with hands-on experience in analog/digital circuit design, PCB development, component selection, sensor interfaces, and multi-channel DAQ systems. Proficient in hardware bring-up and benchtop validation using oscilloscopes, VNAs, and signal generators, with experience working with digital interfaces including I2C and SPI. Skilled in automated data acquisition, analysis, and test development using Python and MATLAB.

Education

Northwestern University (NU) – McCormick School of Engineering | Evanston, IL | BS in Electrical Engineering |

Expected Graduation: June 2028 | GPA: 3.7 / 4.0

Dean's List (Winter Quarter 2024); High Honor Student (Fall Quarter 2025, Spring 2026)

Experience

pH Sensing System | Rogers Lab, Querrey Simpson Institute for Bioelectronics, NU

Summer Undergraduate Research Grant (SURG) Recipient | Evanston, IL

Jan 2026 - present

- Architected an automated multi-channel DAQ platform utilizing an ESP32, precision ADC, and ThingSpeak cloud logging; captured and processed 90k+ sensor telemetry points across broad pH/temperature ranges to characterize long-term sensor drift.
- Designed and executed controlled pH-exposure and temperature-perturbation experiments, quantifying calibration-slope shifts from -40.4 to -31.7 mV/pH and identifying temperature and prior-exposure dependent changes in sensor response
- Analyzed time-series data in Python with 10 statistical and signal-processing methods evaluated performance with RMSE and R^2 and reduced short-term sample-to-sample signal variation by $\sim 84\%$

Projects

FloraLogic Watering | Design, Thinking, and Communication 2, NU | Evanston, IL

Apr – Jun 2026

- Designed analog front-end and signal conditioning circuits to interface analog capacitive moisture sensors with the ESP32 ADC, implementing RC low-pass filtering and calibration to mitigate ADC non-linearity.
- Designed low-side MOSFET drive circuitry with flyback diode protection to actuate a 12V solenoid valve safely from 3.3V GPIO logic.
- Performed end-to-end hardware bring-up and power rail validation, characterizing switching transients, sensor response, and system behavior during sustained field operation.

Fishing Frenzy | Design, Thinking, and Communication 1, NU | Evanston, IL

Sep – Dec 2025

- Engineered a modular assistive game system integrating switch-activated vibration motors and control electronics to provide tactile feedback, and support simultaneous gameplay for up to 10 players.
- Developed and evaluated 3 design iterations through CAD, prototyping, a client interview, user testing with North Center, and peers, refining interaction mechanisms, hardware layout, and accessibility based on user feedback.

Technical Skills

- **Hardware & Lab:** Schematic Capture & PCB Design (KiCad), Rapid Prototyping, Soldering Rework, Microcontrollers, ADCs, digital interfaces (I2C/SPI/UART), Multimeters & Oscilloscopes, Function generator
- **Programming & Scripting:** Python (PyVISA, NumPy, SciPy, Pandas), MATLAB, C/C++ (Arduino/ESP-IDF basics), Racket
- **Tools & Platforms:** KiCad, Git, ThingSpeak (IoT Cloud)